

Modeling the Mind in Space: Integrating Cognitive Science and Phenomenology for Human-Centered Architectural Design

Foundational Lenses: From Husserlian Intentionality to Enactive Embodiment

The interdisciplinary field at the intersection of cognitive science and phenomenological architecture is anchored in a profound shift in how we understand human experience. This shift moves away from abstract, disembodied models of consciousness toward a dynamic, embodied, and relational framework where the built environment is not merely a backdrop but an active participant in shaping lived reality [2](#) [51](#). The theoretical foundations of this new paradigm draw heavily from three key phenomenological traditions—Husserlian, Heideggerian, and Merleau-Pontyan—which provide the philosophical scaffolding for contemporary cognitive theories and their application in architectural design. These traditions collectively argue against viewing the mind as a passive processor of information, instead positing that perception, cognition, and action are fundamentally intertwined through bodily engagement with the world [23](#).

The foundation of this inquiry is rooted in the work of Edmund Husserl, whose concept of intentionality—that consciousness is always consciousness *of* something—provides a starting point for understanding our directedness toward the world [20](#). This principle asserts that our mental states are inherently relational, pointing toward objects, places, and experiences. In an architectural context, this means a space is never perceived as a neutral container; rather, it is always perceived *as* something—a cathedral, a home, a prison—because consciousness is always already engaged with its contents. This perspective forms the bedrock of architectural phenomenology, which seeks to describe the essential structures of spatial experience [14](#). However, it was Martin Heidegger who radicalized this focus, shifting from the subject-object relationship of Husserl to his concept of "being-in-the-world" [20](#). Heidegger argued that human existence is fundamentally situated and relational, pre-reflective and practical. We are not detached observers looking *at* the world; we are embedded within it, our being defined by our everyday dealings with tools, spaces, and others. This has direct implications for architecture, offering a powerful critique of modernist designs that prioritize abstract

form over lived, practical engagement, thereby severing the deep connection between human existence and its physical setting [11](#) [20](#) . The architect's task, from this viewpoint, is not to create autonomous aesthetic objects but to disclose the world in a way that supports authentic modes of being.

Maurice Merleau-Ponty's philosophy provides the most crucial and frequently cited lens for bridging phenomenology and contemporary cognitive science, directly addressing what some scholars term a "blind spot" in the contemporary architectural design process [11](#) [12](#) . His central contribution is the concept of embodiment, articulated through the ideas of the "lived body" (*le corps propre*) and "lived space" [18](#) . For Merleau-Ponty, the body is not an object in the world, like any other, but the very medium through which we perceive and interact with it [13](#) . It is the site where subjectivity and objectivity co-penetrate, the primary instrument of our engagement with the world [18](#) . This embodied perspective is instrumental in helping identify a "blind spot" in contemporary architecture design process [11](#) . Our experience of architectural space is not a purely visual or cognitive affair; it is enacted through the full sensorimotor system. As he wrote in *Phenomenology of Perception*, the life of consciousness—cognitive, perceptual, or affective—is subtended by a pre-reflective bodily life [16](#) . This idea finds strong support in the cognitive science principle of embodied cognition, which posits that our thoughts, feelings, and perceptions are shaped by our physical interaction with the environment [2](#) . Merleau-Ponty's work suggests that the value of people's experience of the world gained through their immediate bodily engagement remains paramount, a sentiment echoed in architectural practices that emphasize movement and sensory richness [13](#) .

This confluence of phenomenological insights and cognitive science has given rise to the enactive approach to cognition, which serves as a powerful bridge between philosophy and empirical research [30](#) . Enactivism posits that cognition is not a process of passively constructing internal representations of the world, but an active, skillful engagement where perception and action are mutually constitutive [23](#) . Perception is not a one-way flow of data from the world to the brain; it is an exploratory activity where the agent actively probes the environment to uncover its structure [23](#) . This view resonates deeply with the ecological psychology of J.J. Gibson, who introduced the concept of affordances—the possibilities for action that an environment offers relative to an organism's capabilities (e.g., a surface affords walking on if it is firm and horizontal) [23](#) . The enactive perspective also draws on Heidegger's "being-in-the-world" and Husserl's focus on lived experience, framing cognition as a process of sense-making that emerges from the continuous coupling between an agent and its environment [23](#) [29](#) . A core tenet of enactivism is the principle that "the world is its own best model," suggesting that the

most reliable information about the world is found directly within it through interaction, rather than in any finite internal representation ²³. This leads to the concept of the "intentional arc," a feedback loop where an agent's actions shape their sensory input, which in turn informs more refined subsequent actions, revealing new aspects of the environment ²³. This framework has been applied to diverse fields, including robotics, education, and human-computer interaction, providing a robust theoretical basis for designing interactive systems and environments that support meaningful engagement ²⁵.

Computational Formalisms: Modeling the Dynamics of Architectural Experience

While phenomenological traditions provide rich descriptions of lived experience, and enactivism offers a compelling philosophical framework, the integration of cognitive science requires a formal, mathematical language to move from qualitative insight to testable hypotheses. The most significant recent development in this area is the emergence of computational neurophenomenology, particularly through the use of predictive processing frameworks like Active Inference and the Free Energy Principle ²⁹ ³³. This approach, initiated by Francisco Varela, aims to establish "mutual constraints" between first-person subjective reports and third-person objective neurophysiological data, creating a rigorous methodology for investigating the nature of consciousness itself ³³. This "deep computational neurophenomenology" (DCNPh) specifically targets the modeling of the "how" of experience—the dynamic processes and structures of consciousness—as opposed to the traditional focus on the "what" (i.e., neural correlates of specific content) ²¹ ³³.

The core of this computational turn lies in treating the brain as a Bayesian inference engine that operates to minimize a quantity known as free energy, which is a proxy for surprise or prediction error ⁴¹ ⁴². According to the Free Energy Principle, all living systems must maintain their integrity by resisting a tendency to collapse into equilibrium with their environment. They achieve this by maintaining a generative model of their environment and minimizing surprise by either updating this model (through learning) or acting upon the world to make it conform to the model's predictions (Active Inference) ⁴². This framework provides a unified, dynamical explanation for perception, action, and learning. In the context of architectural experience, this suggests that our encounter with a space is not a passive reception of sensory data but an active process of generating predictions about the environment and continuously testing them through movement and

exploration 37. A space that is easy to navigate and understand allows for accurate predictions, leading to a state of low free energy and a sense of ease. Conversely, a confusing or disorienting space generates high prediction errors, demanding constant inferential effort and resulting in stress or anxiety 1.

Deep computational neurophenomenology leverages these principles to build formal models that can simultaneously account for both subjective experience and neurophysiological activity 40. The methodology involves a tight feedback loop, described as a "triple braid," which iteratively refines the model by constraining it with data from multiple domains 33. First, trained participants engage in disciplined phenomenological inquiry, using techniques like "dereification" to shift from a natural, pre-reflective immersion in objects to a reflective awareness of the underlying processes of experience itself 33. Their detailed, first-person reports about the "how" of their experience—such as fluctuations in attention, affective tone, or pre-reflective self-awareness—are used to constrain the extrinsic dynamics of the computational model. Second, third-person neurophysiological data, such as EEG recordings, are used to constrain the intrinsic dynamics of the model, linking subjective states to measurable brain activity 33. Finally, the resulting integrated model generates novel predictions that can be tested against new data in either domain, creating a meta-Bayesian cycle of refinement 33. This approach allows researchers to formalize and investigate subtle aspects of conscious experience, such as minimal phenomenal experience (MPE), which arises when an agent achieves low free energy through self-directed awareness and modulation of its generative model 50 55. This computational formalism thus bridges the gap between phenomenological descriptions and physiological explanations, offering a powerful tool for empirically investigating how architectural spaces might induce specific types of perceptual or affective states through their influence on sensorimotor dynamics 35 37.

Concept	Description	Relevance to Architecture
Intentionality	Husserl's concept that consciousness is always directed toward an object or aspect of the world. 20	Explains why spaces are perceived as having qualities (e.g., welcoming, oppressive) rather than being neutral containers.
Being-in-the-World	Heidegger's idea that human existence is fundamentally situated and relational, prior to reflection. 20	Critiques abstract architectural forms and emphasizes designing for practical, everyday engagement and use. 11
Embodiment / Lived Body	Merleau-Ponty's thesis that the body is the primary medium through which we perceive and act in the world, not just an object in it. 13 18	Grounds architectural experience in the full sensorimotor system, explaining how geometry, texture, and light affect emotion and stress. 1 2
Enaction	An approach positing that cognition is an active, skillful engagement where perception and action are mutually constitutive. 23	Shifts the architect's role from designer of static objects to choreographer of sensorimotor interactions and exploratory activities. 35
Affordances	The properties of an environment that offer opportunities for action to an organism (e.g., graspable, climbable). 23	Provides a framework for designing spaces based on the user's physical capabilities and intended activities.
Active Inference	A computational process where an agent acts on the world to make it match its internal predictions, thereby minimizing uncertainty. 38 42	Models architectural experience as a continuous interplay of prediction and correction, explaining how spaces guide behavior and reduce cognitive load. 37

Empirical Frontiers: Neuroarchitecture and Human-Building Interaction

The theoretical frameworks derived from phenomenology and cognitive science are being translated into tangible empirical research through two primary fields: neuroarchitecture and Human-Building Interaction (HBI). While both aim to create more human-centered environments, they differ in their methodologies and scales of application.

Neuroarchitecture employs neuroscientific techniques to understand the neural mechanisms underlying our experience of space, while HBI focuses on developing intelligent building systems that can adapt to occupants' needs in real time. Together, they represent a growing body of evidence that the built environment has a direct and measurable impact on human well-being, performance, and behavior [4](#) [28](#) .

Neuroarchitecture represents a direct attempt to quantify the qualitative. Using tools like functional magnetic resonance imaging (fMRI), researchers investigate the neural correlates of spatial perception, navigation, and aesthetic appreciation [4](#) [5](#) . Findings from this field indicate that elements of the built environment, such as geometry, curvature, and lighting, directly influence brain activity in regions associated with

emotion, stress, and memory [1](#) . This provides empirical validation for the long-held intuition that spaces can evoke specific psychological and physiological responses. Research in this area draws upon predictive coding and multisensory integration theories, suggesting that our experience of space is an integrated, predictive process rather than a simple sum of sensory inputs [39](#) . For instance, studies have explored how architectural features can modulate behavior through automated sensorimotor responses [3](#) . Other innovative approaches, such as contemplative neuroaesthetics, aim to develop frameworks for designing spaces that can support meditative and mindful states by leveraging sensorimotor dynamics [32](#) [45](#) . These studies contribute to a deeper understanding of the interactions between people and their environment, moving beyond aesthetics to consider the fundamental cognitive and affective processes involved [4](#) .

In contrast, Human-Building Interaction (HBI) is a more applied, systems-level discipline focused on creating responsive and adaptive built environments [28](#) . HBI is defined as an interdisciplinary field aiming to understand how built environments affect human outcomes and experiences, and how humans interact with, adapt to, and affect the built environment and its systems [28](#) . The goal is to enable buildings that can learn, adapt, and evolve at various scales—from individual rooms to entire cities—to improve quality of life and optimize resource usage [28](#) . This field relies on a suite of enabling technologies, including real-time locating systems for tracking assets and people, advanced sensors like Doppler radar for detecting physiological parameters such as heart rate and respiration, and a range of AI techniques for predictive modeling and control [28](#) . Supervised learning, unsupervised learning, and reinforcement learning are all employed to analyze data and make intelligent decisions about building operations [28](#) .

Several real-world projects exemplify the application of HBI principles. The Cortellucci Vaughan Hospital in Ontario, Canada, which opened in 2021, uses a smart technology grid and real-time locating systems to manage logistics and staff workflows efficiently [28](#) . Similarly, the EDGE building in Amsterdam utilizes an integrated system that knows workers' schedules to assign parking spots and customize workstation conditions like lighting and temperature to personal preferences [28](#) . These examples demonstrate the successful implementation of collecting and responding to human data to enhance both operational efficiency and occupant comfort. However, the deployment of such systems is not without significant challenges. A critical issue highlighted in the literature is algorithmic bias. One study noted that a depth-imaging occupancy sensor misestimated occupancy based on hair type due to differences in infrared light absorption, illustrating how seemingly neutral technologies can produce inequitable outcomes [28](#) . This underscores the necessity of incorporating core ethical principles—Equity and Inclusion,

Privacy and Security, and Sustainability and Resilience—into every aspect of HBI research and design [28](#). Proposed frameworks like the EU's "AI Ethics by Design" model advocate for integrating ethical principles directly into the development lifecycle of these technologies to ensure they serve human needs responsibly [7](#) [10](#). The convergence of these fields suggests a future where architectural spaces are not only designed based on neuroscientific insights but are also equipped with intelligent systems that continuously adapt to the real-time experiential and physiological states of their inhabitants.

AI-Informed Design: Collaborative Co-Creation and Socioenactive Systems

Artificial intelligence is rapidly evolving from a mere design tool into a potential collaborator in the creation of human-centered environments, giving rise to new paradigms of co-creation and socioenactive systems. This evolution is marked by a shift from using AI for simple optimization tasks to envisioning AI as an integral part of a larger, distributed cognitive system that includes humans, artifacts, and the environment [25](#) [27](#). Generative AI, in particular, is positioned as a transformative force in architectural practice, capable of producing and assessing design options based on vast datasets of quantitative and qualitative information [6](#) [8](#). This technology enables the rapid generation of early-stage sustainable designs, incorporating criteria such as energy efficiency, material sourcing, and even user feedback, thus accelerating the design iteration process [9](#).

A mixed-methods study conducted in 2025 on designing urban furniture for public parks provides a concrete example of this collaborative workflow [27](#). In this research, AI applications were used to generate preliminary designs, which were then refined and guided by human designers to ensure they met functional, aesthetic, and user-centered requirements. The proposed designs integrated sustainable materials like recycled wood with smart features such as solar-powered lighting and interactive screens [27](#). Public opinion polling revealed strong user support for these integrated technologies, with over 89% of respondents agreeing on the importance of sustainable materials and 84% on the impact of well-designed furniture on visitor happiness [27](#). This case illustrates a symbiotic relationship: AI handles the computationally intensive tasks of data analysis and initial ideation, while human expertise remains essential for contextual judgment, functional evaluation, and ensuring the final product aligns with nuanced human values and societal goals [27](#). However, the study also identified several limitations of current AI

technologies, including shortcomings in evaluating functional usability and considering the full lifecycle impact of a design, reinforcing the need for continued human oversight [27](#).

Beyond its role as a generative tool, AI is being conceptualized within socioenactive frameworks that treat intelligence as a distributed property emerging from the continuous interaction between agents and their environment [25](#). The concept of 'socioenactive experience' proposes a tripartite coupling of Social, Physical, and Digital elements, where 'smartness' is an emergent property of this inseparable system [25](#). In this view, an intelligent building is not just a collection of algorithms but an adaptive entity co-created through its interactions with occupants and the physical space [25](#). This perspective is mirrored in the development of new AI paradigms like the Phenomenological Experiential Intelligence (PEI) framework [24](#). PEI is proposed as a new paradigm for AI grounded in phenomenology, enactive cognition, and distributed cognition. Its goal is to create AI that is truly embodied, context-aware, and ethically aligned with human values by positioning lived experience at the center of its learning process [24](#). The PEI architecture is structured around three layers: Narrative Context, Micro-Phenomenology, and Psychophysiology, allowing the AI to construct a rich, temporally layered understanding of a user's state by combining macro-contextual information, fine-grained experiential reports, and direct physiological measurements [24](#).

This enactive approach to AI critiques the current mainstream reliance on large language models (LLMs) and other systems based on a representational theory of mind. Such AI lacks grounding in physical reality because it processes text without the sensorimotor feedback loops that tether human understanding to the world [24](#). This disembodiment leads to well-known issues like "hallucinations"—confidently generating false information—and failures in commonsense reasoning, as the models manipulate symbols without genuine comprehension of their referents [24](#). By contrast, PEI aims to create an AI that learns about a user's state not by retrieving static data but by actively engaging with them, eliciting responses, and interpreting those responses within the specific narrative context of the interaction. This mirrors Francisco Varela's idea that knowledge is enacted through participation, not passively retrieved [24](#). This vision of AI as a cognitive partner within a socioenactive system points toward a future where intelligent environments do not simply respond to commands but participate in a continuous dialogue with their inhabitants, co-creating a shared, meaningful experience. However, this raises profound ethical questions about agency, responsibility, and the potential for manipulation, necessitating the development of robust ethical guardrails before such systems become widespread [7](#) [20](#).

Emerging Design Principles for Human-Centered Environments

The synthesis of cognitive science and phenomenological architecture yields a set of guiding principles for designing spaces that are genuinely human-centered, prioritizing the experiential qualities of consciousness. These principles move beyond conventional functional and aesthetic criteria to address the fundamental ways in which humans perceive, feel, and act within their surroundings. They are informed by a deep understanding of embodied cognition, predictive processing, and the enactive nature of experience, and they point toward a future where architecture actively contributes to human flourishing. Key among these principles are the embrace of embodied cognition, the design for sensorimotor engagement, the prioritization of predictive and anticipatory dynamics, and the operationalization of meaning and purpose.

First and foremost is the principle of embracing **embodied cognition**, which dictates that design must be predicated on the understanding that our thoughts, emotions, and perceptions are shaped by our physical interaction with the environment [2](#) [51](#). This requires a holistic approach that considers the full spectrum of sensory-motor engagement, not just visual aesthetics. Research confirms the significance of this multisensory approach; a study on park furniture design found that users strongly preferred colors derived from nature, as they contributed to a feeling of vitality and activity, and that overall design significantly impacts psychological safety and happiness [27](#). This principle calls for architects to think beyond sight and sound, attending to factors like thermal comfort, material textures, air quality, and acoustics as integral components of the architectural experience [39](#). The goal is to create environments that feel coherent and supportive at a visceral level, aligning with the body's natural rhythms and expectations.

Second is the principle of **designing for sensorimotor engagement**. Drawing from the enactive perspective, this principle reframes the architect's role from a creator of static objects to a choreographer of movement and interaction [23](#). Instead of designing spaces as backdrops for human activity, the focus shifts to designing the patterns of interaction themselves. This aligns with the concept of the "architectural promenade," where continuous movement shapes the embodied experience, a technique adapted in video game design through the use of long takes to create immersive, uninterrupted journeys through virtual spaces [26](#). By designing paths, thresholds, and vistas that invite exploration and reveal themselves progressively, architects can foster a deeper, more meaningful connection between the inhabitant and the place. This approach treats the

user not as a passive observer but as an active participant whose movements are essential to the unfolding of the architectural experience [35](#).

Third, a critical principle derived from computational neuroscience is to **prioritize predictive and anticipatory dynamics**. Architectural experience arises from a continuous interplay between prediction and correction, a process modeled by frameworks like Active Inference [37](#) [38](#). Spaces that allow for predictable patterns of movement, clear legibility, and consistent structural logic reduce cognitive load and promote a sense of ease and coherence. Conversely, environments that are overly complex or unpredictable can lead to heightened stress and anxiety as the brain struggles to form accurate models of the situation [1](#). Designing for anticipation involves creating spaces that guide the user intuitively, leveraging established sensorimotor contingencies so that perception becomes effortless and seamless. This principle suggests that clarity and intelligibility are not just stylistic choices but are fundamental to creating positive experiential outcomes.

Finally, emerging frameworks suggest a future principle of **operationalizing meaning and purpose**. The Neuro-Existential Architecture System (NEAS) proposes that meaning is not an abstract, psychological add-on but a thermodynamic necessity for optimizing the brain's predictive dynamics [31](#) [46](#). According to this view, the brain's "Spirit" or "Logos" functions to maintain viability at the edge of chaos, preventing pathological states like fanaticism or depression [31](#). This implies that architecture could be designed not just for comfort or efficiency, but to actively support users in finding purpose and navigating existential challenges. Such a space would function as a "technology of enactment," a concept that redefines spiritual practices as training regimes for the brain to maintain optimal information processing and viability [31](#). While highly speculative, this line of inquiry pushes the boundaries of architectural design toward a more profound role in supporting human well-being, addressing not just our physical and cognitive needs but also our existential ones. Together, these principles provide a comprehensive guide for a new generation of architects seeking to create built environments that are deeply resonant with the totality of human experience.

Synthesis and Future Directions: Toward an Experiential Architecture

The convergence of cognitive science and phenomenological architecture is catalyzing a significant evolution in design philosophy and practice, moving toward a more profound integration of human experience at the core of the built environment. This synthesis, spanning from foundational philosophical inquiries to cutting-edge computational models and empirical applications, establishes a new paradigm where architecture is understood not as a static shell but as an active, participatory force in the co-creation of consciousness. The journey from Husserlian intentionality and Heideggerian being-in-the-world to the formal languages of deep computational neurophenomenology marks a progression from descriptive phenomenology to a rigorous, interdisciplinary science of experience. This report has systematically investigated these dimensions, revealing a field driven by the central goal of creating human-centered environments that are cognitively intelligent, emotionally resonant, and existentially supportive.

The foundational layer of this new paradigm rests on the enactive synthesis, which grounds cognition in the body's continuous interaction with the world [23](#). This perspective, heavily influenced by Merleau-Ponty's emphasis on the lived body, provides the rationale for moving beyond purely formal or representational design towards architectures that choreograph sensorimotor engagement [11](#) [13](#). This shift is not merely theoretical; it is being made tangible through empirical fields like neuroarchitecture and Human-Building Interaction (HBI). Neuroarchitecture provides the "why" by using neuroscience to link environmental features to measurable changes in brain activity related to emotion and stress [1](#) [4](#). HBI provides the "how," using AI and sensing technologies to create responsive buildings that adapt to occupant needs in real time, as demonstrated by projects like the Cortellucci Vaughan Hospital and the EDGE building [28](#). Together, they form a powerful feedback loop: neuroscientific insights inform better HBI systems, and data from HBI systems can validate and refine neuroscientific models.

The most forward-looking aspect of this convergence is the role of artificial intelligence. Generative AI is emerging as a collaborative partner in the design process, augmenting human creativity with the ability to process vast datasets and explore complex design spaces for sustainability and user preference [6](#) [9](#). More radically, proposals for new AI paradigms like Phenomenological Experiential Intelligence (PEI) and socioenactive systems challenge the very nature of machine intelligence, advocating for architectures grounded in embodied, context-aware, and distributed cognition [24](#) [25](#). This moves AI from a tool for optimization to a potential participant in a shared, co-created experience.

However, this trajectory is fraught with unresolved questions and ethical challenges. The hard problem of consciousness remains, leaving open whether an AI operating on these principles could ever possess genuine phenomenal experience ²⁹. Furthermore, the risk of algorithmic bias, as seen in occupancy sensors, highlights the critical need for embedding equity and inclusion as non-negotiable design principles from the outset ⁷ ²⁸.

Looking ahead, the path forward requires a concerted effort to bridge existing gaps. There is a pressing need for larger-scale, mixed-methods research that integrates the fine-grained, first-person data of phenomenology with the broad, third-person data of HBI and neuroimaging ²⁸ ³⁶. This will be essential for translating laboratory findings into robust, generalizable design guidelines applicable across diverse cultural and environmental contexts. Moreover, the development of sophisticated AI necessitates the parallel development of equally sophisticated ethical frameworks. The concept of "human-artificial co-consciousness" demands a new level of scrutiny regarding agency, accountability, and the potential for manipulation in increasingly intelligent environments ²⁰. The ultimate promise of this interdisciplinary endeavor is to realize a truly human-centered architecture—one that honors the full complexity of our embodied, cognitive, and conscious lives. The actionable steps for practitioners are clear: adopt an enactive mindset, integrate mixed-methods research, collaborate with AI as a partner, and embed ethics proactively. The goal is no longer simply to design for human needs but to design for human flourishing, creating spaces that are not only functional and sustainable but also deeply attuned to the art of being in the world.

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